Process Verification

Lab

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Contents

Chapter 1

Design Overview

1.1 Controller

A dual-master memory controller. Master 0 (CPU) has fixed priority over Master 1 (DMA). Synchronous writes on clock, and combinational reads.

1.1.1 Parameters

There are two parameters to this dut:

- `ADDR_WIDTH` : address size.
- `DATA_WIDTH` : data width.

1.1.2 Interface

1.1.3 Functional Description

- **Reads**: 0 clock cycles (combinational —immediate once address & grant are stable).
- **Writes**: 1 clock cycle (synchronous —update happens on next rising edge when grant & we are true).

Signal	Direction	Width	Description
`clk`	Input	1	Clock signal
`rst_n`	Input	1	Active-low synchronous reset
`addr0`	Input	ADDR_WIDTH	Master 0 address
`wdata0`	Input	DATA_WIDTH	Master 0 data
`rdata0`	Output	DATA_WIDTH	Master 0 data
`we0`	Input	1	Master 0 write enable
`req0`	Input	1	Master 0 read request
`gnt0`	Output	1	Master 0 grant
`addr1`	Input	ADDR_WIDTH	Master 1 address
`wdata1`	Input	DATA_WIDTH	Master 1 data
`rdata1`	Output	DATA_WIDTH	Master 1 data
`we1`	Input	1	Master 1 write enable
`req1`	Input	1	Master 1 read request
`gnt1`	Output	1	Master 1 grant

Table 1.1: Controller signals

Condition	Behavior
Only one master valid	That master is serviced immediately
Both masters valid	Master 0 is serviced (priority). Master 1 is not granted until Master 0 de-asserts `re`
No master valid	Controller idle

Table 1.2: Controller Behavior

Appendix A

Appendix Chapter

A.1 Use of tools

- Visual Studio Code: systemverilog IDE.
- QuestaSim: CLI tester.
- GTKwave: wave visualizer.

A.2 Use of time

A total of h distributed as follows.

- Understanding the problem: 1h.
- Report: 1h.

A.3 Use of AI

- Gemini (Pro): sentence enhancing, bug fixer